

# The Early Net: Catching Stroke in Older Adults with Proactive, Multifactorial Monitoring

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## Executive Summary

Research suggests that stroke is related to a variety of factors, involving BP, medical history of TIA and HT. Further investigations found even more factors related to stroke risk, and mandate a dual approach: combining evidence-based screening thresholds with tailored health technology to flag high-risk individuals and support their ongoing health management. This strategy not only improves early detection but also ensures health resources are used wisely. These findings fill gaps in current programs by making screening more practical on stroke prevention in the aged population.

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# 1 Literature Review

## 1.1 Problem Statement

Stroke occurs when blood supply to a part of brain is cut off, causing brain tissue damage (Centers for Disease Control and Prevention, 2024). It is well-known that stroke is related to a number of factors.

For example, hypertension (HT), or high blood pressure (BP), is a leading cause of stroke. It is found that poor treatment of HT often results in higher stroke risk (Han et al., 2017). Having a TIA (transient ischemic attack) also means a higher risk of developing stroke (Centers for Disease Control and Prevention, 2024). In Hong Kong, stroke was the 4th most common cause of death in Hong Kong in 2024 (Centre for Health Prevention, 2025), causing 2,911 deaths per year. It is also found that stroke has a higher incidence among those aged over 55 (Boehme et al., 2017).

The newly-introduced long-term monitoring scheme, CDCC (Chronic Disease Co-care) Pilot Scheme (Primary Healthcare Commission, n.d.) provides screening on HT, diabetes and blood lipid to older adults (>45 years), with possible follow-up actions to mediate HT conditions. However, it only aims to screen and treat chronic diseases, and not directly prevent acute occurrences of other diseases (such as stroke) caused by chronic ones.

Also, basing prevention only on these screening results may miss risk caused by other well-known factors of stroke, such as medical history of TIA. In short, this screening of HT is not effective enough to mediate stroke risk in older adults.

As active measures against stroke are not being taken, stroke will continue to be a dominant problem for the aged population.

## 1.2 Significance and rationale

The unfilled gap of CDCC causes stroke risk to go unnoticed – even if screening shows signs of HT or diabetes, treatment from CDCC Pilot Scheme would focus on treating the chronic disease itself, not acute illnesses like stroke caused by them.

If, alongside managing chronic diseases, further estimations of stroke risk can be performed, proactive actions can be taken against stroke. This means a higher rate of stroke being caught early, saving thousands each year.

## 1.3 Objectives

This study aims to integrate proactive BP monitoring alongside monitoring on other factors such as medical history and lifestyle in the CDCC, to prevent stroke in the aged population.

Specifically,

1. In adults aged 45-80, identify high-risk groups with medical conditions or lifestyle factors (e.g., diet or consumption of certain drinks) that could lead to higher stroke risk, therefore defining groups for prioritized stroke risk monitoring.
2. Assess whether baseline treatment status (e.g., use of HT drugs or diabetes treatment) could mediate or reduce correlation between BP values and stroke risk.
3. Explore correlation between recent symptoms and stroke risk.
4. Generate easy-to-apply screening criteria that prioritize individuals for proactive stroke risk monitoring, and report their predictive performance and population yield.
5. Propose health policy and innovative health technology to implement proactive BP monitoring schemes to predict and prevent stroke in the aged population.

## 2 Data analysis

### 2.1 Methodology

A retrospective cohort is used to compare trends in BP (systolic and diastolic) that may lead to stroke, and correlation between symptoms reported at wave preceding stroke to find correlation between occurrence of symptoms and stroke risk.

At baseline, individuals aged 45-80 and free of stroke are included. Only individuals that appeared across 3 or more “waves” (statistical periods) are included.

### 2.2 Data Import

This part processes necessary data to prepare for analysis. The data is downloaded from CHNS (China Health and Nutrition Survey). (<https://chns.cpc.unc.edu/>), licensed under CC0.

All datasets provided are in the SAS format. They are read into Pandas `DataFrames` first for further processing.

**Clean up Data** CHNS uses a set of codes to name their data columns, which are available with the datasets. Individuals have a unique ID assigned to them, which is persistent across waves (years where CHNS is taken place). This means that one can combine individual data across waves for longitudinal study.

For the sake of simplicity, only statistical data collected that are related to cardiovascular and cerebrovascular problems are taken into consideration. The following are factors included.

- Age
- Gender
- Recent symptoms
  - Had a headache in last 4 weeks?
  - Had chest pain in last 4 weeks?
  - Felt exhausted in last 4 weeks?
  - Had bleeding of unknown cause in last 4 weeks?
  - Had weight loss of unknown cause in last 4 weeks?
- Systolic blood pressure (take mean of 3 values)
- Diastolic blood pressure (take mean of 3 values)
- BMI (calculated from height and weight)
  - Height
  - Weight
- Has hypertension?
  - Is the patient taking HT drugs?
- Has diabetes?
- Has myocardiac infarction (MI)?
- Has apoplexy?
- Had a transient ischemic attack (TIA) in the past year?
- Had ever smoked?
  - Still smoking or quitted?
  - Number of cigarettes smoked per day.
- Drinks beer?
  - Number of bottles of beer per week
  - Frequency of drinking beer per week
- Drinks tea?
  - Number of bottles of tea per week

- Frequency of drinking tea per week
- Drinks coffee?
  - Number of bottles of coffee per week
  - Frequency of drinking coffee per week

Frequency of all **\_Freq** columns are encoded as follows:

1. Almost every day
2. 4-5 times a week
3. 2-3 times a week
4. No more than once a week
5. 2-3 times in past 30 days
6. Only once in past 30 days
7. None in past 30 days

**Merge and filter** Merge health data and health examination data. Then filter out individuals:

- Not aged 45-80 at baseline
- Not present in more than 2 “waves”

Remove population that has had a apoplexy (stroke) at baseline.

## 2.3 Hazard ratio of variables

This part will assess correlation between factors recorded in the datasets and stroke risk.

**Strategy** Two rounds of Cox model evaluations are created to predict the hazard ratio of variables.

**First round** A model is fitted from the following variables, each of which is used as a predictor:

- Time\_Yrs - Event follow-up time
- Event - Whether stroke occurred
- Slope\_SBP - SBP slope from start to end of waves
- Slope\_DBP - DBP slope from start to end of waves
- Mean\_SBP - Mean of SBP from start to end of waves
- Mean\_DBP - Mean of DBP from start to end of waves
- Age\_Baseline - Age at baseline t=0
- Variation of BP
  - SD\_SBP - Standard deviation of SBP
  - SD\_DBP - Standard deviation of DBP
  - ARV\_SBP - Average Real Variability of SBP
  - ARV\_DBP - Average Real Variability of DBP
- BMI - BMI
- Has\_Untreated\_HT - 0-1 value whether the individual has untreated HT conditions (i.e., has HT but not taking antihypertensive drugs)
- Has\_Untreated\_Diab - 0-1 value whether the individual has untreated diabetes
- Has\_Treated\_HT - 0-1 value whether the individual has controlled HT conditions (i.e., has HT but not taking antihypertensive drugs)
- Has\_Treated\_Diab - 0-1 value whether the individual has controlled diabetes
- Has\_HT - 0-1 value whether the individual has HT conditions (i.e., has HT but not taking antihypertensive drugs)
- Has\_Diab - 0-1 value whether the individual has diabetes
- Has\_MI - 0-1 value whether the individual has untreated MI
- Had\_TIA - 0-1 value whether the individual had a TIA

- **Still\_Smoking** - 0-1 value whether the individual is still smoking
- Life habit factors
  - **Avg\_Beer\_Bottles\_Day** - Average number of bottles of beer consumed per day.
  - **Avg\_Coffee\_Cups\_Day** - Average number of cups of coffee consumed per day.
  - **Avg\_Tea\_Cups\_Day** - Average number of cups of tea consumed per day.

In this Cox model, the following variables are scaled to ensure better visibility. - Means of SBP and DBP - Scaled by 10 times (i.e., per 10 mmHg increase) - BMI - Scaled by 2 times

**Second round** Correlation between the years of having the following conditions and stroke risk is evaluated separately,

- **TIA\_Years** - Years after TIA before stroke
- **HT\_Years** - Years of having HT before stroke
- **Diab\_Years** - Years of having diabetes before stroke
- **MI\_Years** - Years after having MI before stroke.

All of these variables are calculated as such,

1. Calculate age when stroke occurred (from `stroke_wave - baseline_wave + baseline_age`) **(1)**
2. Find age when condition is diagnosed **(2)**
3. Subtract **(2)** from **(1)** to find years of having the condition before stroke.

**Feature extraction** Extracts features (baseline covariates) from dataset for each individual.

This converts the wave dataset created earlier into a person dataset, containing the same factors but merged together for each individual.

**Helper function for evaluating Cox model** This `evaluate_cox` function takes in the persons `DataFrame`, then fits a Cox model on it.

**First round of Cox model** This round evaluates stroke risk on the entire dataset, with a subset of factors,

- Slope, mean, standard deviation, ARV of SBP and DBP
- Age at baseline
- BMI
- Whether individual has HT or diabetes
- Whether individual has HT or diabetes under treatment or under no treatment
- Whether individual is still smoking regularly
- Consumption of beer, coffee and tea

Table 1: Hazard ratios of covariates and  $p$  values

| Covariate           | coef    | exp(coef) | se(coef) | coef lower 95% | coef upper 95% |
|---------------------|---------|-----------|----------|----------------|----------------|
| <b>Slope of SBP</b> | -0.0132 | 0.9869    | 0.0130   | -0.0387        | 0.0123         |
| <b>Slope of DBP</b> | -0.0262 | 0.9742    | 0.0210   | -0.0673        | 0.0149         |
| <b>Mean of SBP</b>  | 0.0138  | 1.0139    | 0.0020   | 0.0099         | 0.0176         |
| <b>Mean of DBP</b>  | 0.0198  | 1.0200    | 0.0034   | 0.0130         | 0.0265         |
| <b>SD of SBP</b>    | 0.0020  | 1.0020    | 0.0031   | -0.0041        | 0.0081         |
| <b>SD of DBP</b>    | 0.0015  | 1.0015    | 0.0050   | -0.0083        | 0.0113         |
| <b>ARV of SBP</b>   | 0.0052  | 1.0053    | 0.0029   | -0.0005        | 0.0110         |
| <b>ARV of DBP</b>   | 0.0063  | 1.0063    | 0.0045   | -0.0026        | 0.0151         |

|                          |         |        |        |         |        |
|--------------------------|---------|--------|--------|---------|--------|
| Age at Baseline          | 0.0173  | 1.0174 | 0.0025 | 0.0124  | 0.0221 |
| BMI                      | 0.0316  | 1.0321 | 0.0096 | 0.0128  | 0.0503 |
| Has Untreated HT         | 0.3608  | 1.4344 | 0.1917 | -0.0150 | 0.7366 |
| Has Untreated Diab       | -1.1558 | 0.3148 | 1.5989 | -4.2896 | 1.9779 |
| Has Treated HT           | 0.5134  | 1.6710 | 0.1456 | 0.2280  | 0.7988 |
| Has Treated Diab         | 0.0161  | 1.0163 | 0.0064 | 0.0037  | 0.0286 |
| Has HT                   | 0.1927  | 1.2125 | 0.0933 | 0.0097  | 0.3756 |
| Has Diab                 | 0.1426  | 1.1533 | 0.1813 | -0.2128 | 0.4981 |
| Has MI                   | -0.0136 | 0.9865 | 0.2011 | -0.4076 | 0.3805 |
| Still Smoking            | 0.0608  | 1.0627 | 0.0639 | -0.0645 | 0.1861 |
| Avg Beer (Bottles) / Day | 0.0004  | 1.0004 | 0.0209 | -0.0406 | 0.0413 |
| Avg Coffee (Cups) / Day  | 0.0725  | 1.0752 | 0.2120 | -0.3431 | 0.4881 |
| Avg Tea (Cups) / Day     | 0.0114  | 1.0115 | 0.0173 | -0.0225 | 0.0453 |

| Covariate                | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z       |
|--------------------------|---------------------|---------------------|--------|---------|
| Slope of SBP             | 0.9621              | 1.0124              | 0.0000 | -1.0131 |
| Slope of DBP             | 0.9350              | 1.0150              | 0.0000 | -1.2489 |
| Mean of SBP              | 1.0100              | 1.0178              | 0.0000 | 6.9975  |
| Mean of DBP              | 1.0131              | 1.0269              | 0.0000 | 5.7414  |
| SD of SBP                | 0.9959              | 1.0082              | 0.0000 | 0.6426  |
| SD of DBP                | 0.9918              | 1.0114              | 0.0000 | 0.3068  |
| ARV of SBP               | 0.9995              | 1.0110              | 0.0000 | 1.7974  |
| ARV of DBP               | 0.9975              | 1.0152              | 0.0000 | 1.3930  |
| Age at Baseline          | 1.0124              | 1.0224              | 0.0000 | 6.9227  |
| BMI                      | 1.0129              | 1.0516              | 0.0000 | 3.3001  |
| Has Untreated HT         | 0.9851              | 2.0887              | 0.0000 | 1.8816  |
| Has Untreated Diab       | 0.0137              | 7.2274              | 0.0000 | -0.7229 |
| Has Treated HT           | 1.2561              | 2.2228              | 0.0000 | 3.5263  |
| Has Treated Diab         | 1.0037              | 1.0290              | 0.0000 | 2.5386  |
| Has HT                   | 1.0098              | 1.4559              | 0.0000 | 2.0644  |
| Has Diab                 | 0.8083              | 1.6455              | 0.0000 | 0.7865  |
| Has MI                   | 0.6652              | 1.4630              | 0.0000 | -0.0675 |
| Still Smoking            | 0.9375              | 1.2045              | 0.0000 | 0.9509  |
| Avg Beer (Bottles) / Day | 0.9602              | 1.0422              | 0.0000 | 0.0172  |
| Avg Coffee (Cups) / Day  | 0.7096              | 1.6292              | 0.0000 | 0.3420  |
| Avg Tea (Cups) / Day     | 0.9778              | 1.0464              | 0.0000 | 0.6602  |

| Covariate       | $p$    | $-\log_2(p)$ |
|-----------------|--------|--------------|
| Slope of SBP    | 0.3110 | 1.6850       |
| Slope of DBP    | 0.2117 | 2.2399       |
| Mean of SBP     | 0.0000 | 38.4810      |
| Mean of DBP     | 0.0000 | 26.6662      |
| SD of SBP       | 0.5205 | 0.9421       |
| SD of DBP       | 0.7590 | 0.3978       |
| ARV of SBP      | 0.0723 | 3.7904       |
| ARV of DBP      | 0.1636 | 2.6115       |
| Age at Baseline | 0.0000 | 37.7152      |
| BMI             | 0.0010 | 10.0147      |

|                          |        |         |
|--------------------------|--------|---------|
| Has Untreated HT         | 0.0599 | 4.0614  |
| Has Untreated Diab       | 0.4697 | 1.0901  |
| Has Treated HT           | 0.0004 | 11.2127 |
| Has Treated Diab         | 0.0111 | 6.4893  |
| Has HT                   | 0.0390 | 4.6811  |
| Has Diab                 | 0.4316 | 1.2122  |
| Has MI                   | 0.9462 | 0.0798  |
| Still Smoking            | 0.3417 | 1.5494  |
| Avg Beer (Bottles) / Day | 0.9863 | 0.0199  |
| Avg Coffee (Cups) / Day  | 0.7324 | 0.4494  |
| Avg Tea (Cups) / Day     | 0.5091 | 0.9740  |

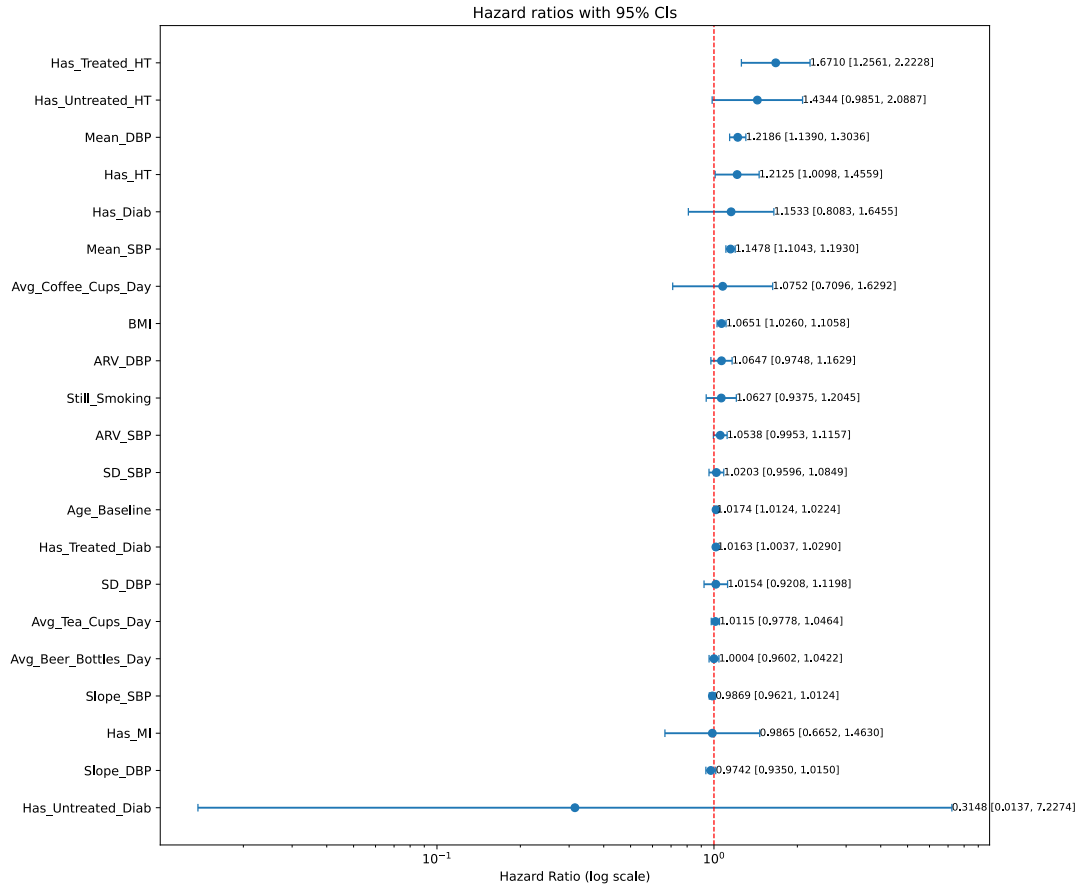


Figure 1: Hazard ratios with 95% CI for first round of Cox model.

#### Note

Had\_TIA was not included in the analysis, as its presence causes errors in the Cox model prediction phase.

**Hazard ratios of variables** From the results,

#### 1. Statistically significant factors ( $p < 0.05$ , $-\log_2 p > 2.9957$ )

1. Mean of DBP ( $\beta = 0.0218$ ,  $p \ll 0.05$ )
2. Mean of SBP ( $\beta = 0.0141$ ,  $p \ll 0.05$ )
3. BMI ( $\beta = 0.0326$ ,  $p = 0.0023$ )
4. Age at baseline ( $\beta = 0.0187$ ,  $p \ll 0.05$ )

5. Whether the individual has HT under treatment ( $\beta = 0.4222$ ,  $p = 0.0054$ )
6. Whether the individual has diabetes under treatment ( $\beta = 0.2228$ ,  $p = 0.0189$ )

These factors agree with existing literature found.

**2. Valid predictors** Valid predictors are defined by predictors with both upper and lower hazard ratios under or over 1.

1. Mean DBP (HR = 1.2441 per 10 mmHg increase)
2. Mean SBP (HR = 1.1509 per 10 mmHg increase)
3. BMI (HR = 1.0673 per 2.0 increase in BMI)
4. Age at baseline (HR = 1.0189 per 1 year increase)
5. Whether the individual has HT under treatment (HT = 1.5253 if so)
6. Whether the individual has diabetes under treatment (HT = 1.0152 if so)

Additionally, slope of SBP and DBP are related to stroke risk, but with low statistical significance ( $-\log_2 p = 2.5287$ ,  $p > 0.05$  respectively).

Meanwhile, variation in SBP and DBP (Standard Deviation and Average Real Variability of SBP and DBP) have little effect on stroke risk ( $\beta < 0.0069$  per mmHg increase), but statistical significance is also low ( $p > 0.05$ ), and their predictors in the Cox model are not valid. Some life habit factors (consumption of beverages), have effect on stroke risk ( $\beta$  for coffee drinking = 0.1991,  $\beta$  for tea drinking = 0.0163). However, their statistical significances are low ( $p > 0.05$ ), and their predictors are not valid in the Cox model. Having cardiovascular diseases, like HT, or having diabetes or MI, does affect the stroke risk, though having low statistical significance ( $p > 0.05$ ).

Therefore, by comparing trends of BP and stroke occurrence, it can be seen that trends of BP (measured by slope of SBP and DBP) has no statistically significant relation with stroke risk. However, it is true that for a generally high BP over the years (especially higher DBP), can lead to a higher risk of stroke.

From results, if a proactive BP monitoring scheme were implemented, it should be prioritized for those who

1. Has higher DBP or SBP
2. Has HT conditions or diabetes conditions (under treatment)
3. Has higher BMI
4. Are older

, as such factors are more correlated with stroke risk than others.

**Second rounds of Cox model** These evaluations tries to find correlation between the years of having a condition and stroke.

Table 5: Hazard ratios of years with TIA and  $p$  values

| Covariate        | coef   | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% |
|------------------|--------|-----------|----------|----------------|----------------|---------------------|
| <b>TIA Years</b> | 0.0017 | 1.0017    | 0.0091   | -0.0162        | 0.0195         | 0.9839              |

| Covariate        | exp(coef) upper 95% | cmp to | z      | p      | $-\log_2(p)$ |
|------------------|---------------------|--------|--------|--------|--------------|
| <b>TIA Years</b> | 1.0197              | 0.0000 | 0.1836 | 0.8543 | 0.2272       |



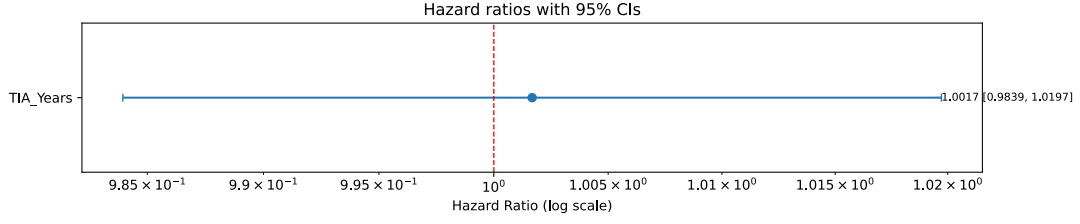


Figure 2: Hazard ratio with 95% CI for years with TIA

Table 8: Hazard ratios of years with HT and  $p$  values

| Covariate | coef    | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% |
|-----------|---------|-----------|----------|----------------|----------------|---------------------|
| HT_Years  | -0.0006 | 0.9994    | 0.0022   | -0.0050        | 0.0038         | 0.9950              |

| Covariate | exp(coef) upper 95% | cmp to | z       | p      | $-\log_2(p)$ |
|-----------|---------------------|--------|---------|--------|--------------|
| HT_Years  | 1.0038              | 0.0000 | -0.2675 | 0.7891 | 0.3418       |

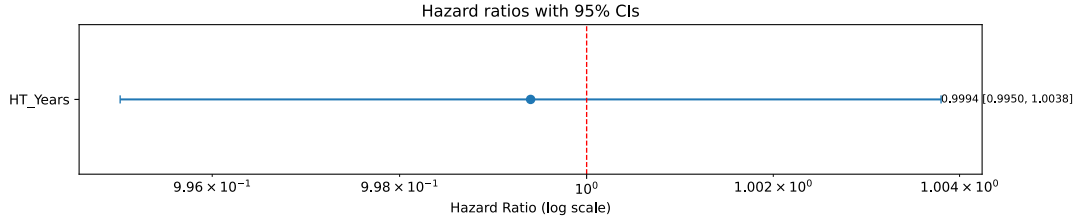


Figure 3: Hazard ratio with 95% CI for years with HT

Table 11: Hazard ratios of years with Diabetes and  $p$  values

| Covariate  | coef    | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% |
|------------|---------|-----------|----------|----------------|----------------|---------------------|
| Diab_Years | -0.0033 | 0.9967    | 0.0040   | -0.0110        | 0.0045         | 0.9890              |

| Covariate  | exp(coef) upper 95% | cmp to | z       | p      | $-\log_2(p)$ |
|------------|---------------------|--------|---------|--------|--------------|
| Diab_Years | 1.0045              | 0.0000 | -0.8281 | 0.4076 | 1.2948       |

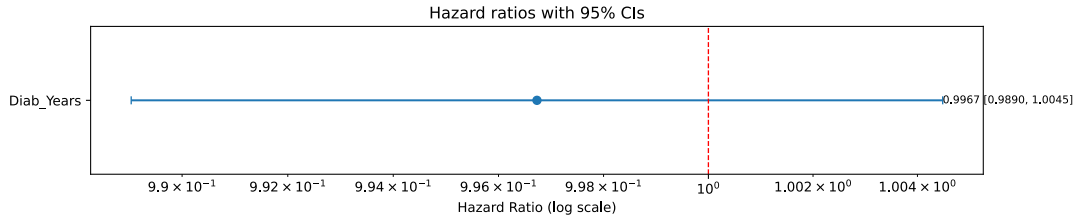


Figure 4: Hazard ratio with 95% CI for years with diabetes

Table 14: Hazard ratios of years with MI and  $p$  values

| Covariate | coef    | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% |
|-----------|---------|-----------|----------|----------------|----------------|---------------------|
| MI_Years  | -0.0014 | 0.9986    | 0.0084   | -0.0179        | 0.0152         | 0.9823              |

| Covariate | exp(coef) upper 95% | cmp to | z       | p      | $-\log_2(p)$ |
|-----------|---------------------|--------|---------|--------|--------------|
| MI_Years  | 1.0153              | 0.0000 | -0.1605 | 0.8725 | 0.1968       |

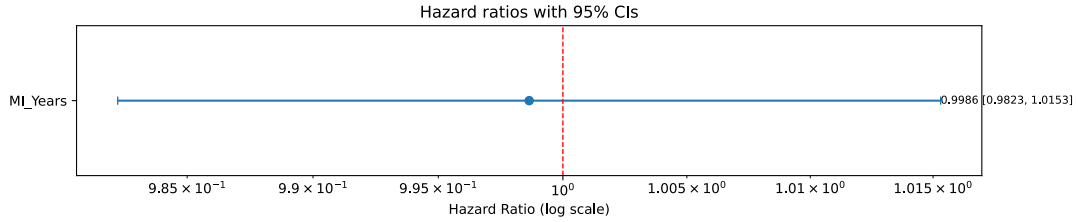


Figure 5: Hazard ratio with 95% CI for years with MI

From the results, none of the years of having the conditions (HT, MI, TIA, Diabetes) shows statistically significant correlation with difference in stroke risk. They are also not valid predictors in the Cox model.

## 2.4 Assess effectiveness of HT and diabetes treatments

Since data is available on individuals receiving HT and diabetes treatments, it is possible to assess whether such treatments mediate or reduce risks of stroke.

From results in first round of Cox model evaluation,

**1. HT under treatment is still associated with higher stroke risk.** HT may contribute to a higher stroke risk. Having HT under treatment, however, does not eliminate this higher risk. Results show that having HT under treatment has around 52.53% higher stroke risk than others. On the other hand, predictors based on untreated HT and HT alone are not valid, nor statistically significant.

As there is no statistically significant correlation between untreated HT for stroke risk to compare with that between treated HT with stroke risk, it is unable to verify findings in Han et al., 2017, which is untreated HT contributes to higher stroke risk compared to treated HT.

**2. Diabetes under treatment is still associated with higher stroke risk.** Diabetes may be associated with a higher stroke risk. However, diabetes under treatment of drugs does not eliminate this risk. Results show that diabetes under treatment has around 1.5% higher stroke risk than others. On the other hand, predictors for untreated diabetes and diabetes alone are not valid, nor statistically significant.

Therefore, from the results, treatment of HT and diabetes does not mediate or reduce stroke risk. Monitoring and stroke prevention strategies should be implemented primarily for population with such conditions, even if they are under treatment.

## 2.5 Recent symptoms and short-term stroke risk

This part will assess correlation between recent occurrence of symptoms and stroke events after them.

Since previously, whether having TIA affects stroke risk was unknown due to it causing errors in the Cox model, this condition is brought down to this part to analyze again. Others like having MI are also brought down.

**Strategy** For each person, find stroke event, and look for symptoms in all waves before it. This creates a  $2 \times 2$  contingency table like such,

|                  | Symptom | No Symptom |
|------------------|---------|------------|
| <b>Stroke</b>    | $a$     | $b$        |
| <b>No stroke</b> | $c$     | $d$        |

$a$  = stroke with symptom,  $b$  = stroke without symptom,  $c$  = symptom but no stroke,  $d$  = no symptom and no stroke.

A Fisher's exact test is run on this contingency table to produce odds ratio and a  $p$ -value. Higher odds ratio (odds  $\gg 1$ ) means that the symptom is much more likely to occur before stroke. Lower  $p$  ( $p < 0.05$ ) means such a result is statistically significant.

Also, a Kaplan-Meier model was fit with population with symptom and those without. Such a model estimates survival probability of population with variation of independent variable. Define survival as being free of stroke. A survival function (probability of being stroke-free) for both cases where the symptom occurred and did not occur were calculated and contrasted.

**Feature extraction** This part calculates values for contingency table by counting  $a$ ,  $b$ ,  $c$  and  $d$  values as mentioned earlier, for each symptom in,

- **Headache\_Last4** - Having a headache in last 4 weeks
- **Chestpain\_Last4** - Having a chest pain in last 4 weeks
- **Exhaustion\_Last4** - Feeling exhaustion in last 4 weeks
- **Unknown\_Bleed\_Last4** - Had unknown bleeding in last 4 weeks
- **Unknown\_WghtLoss\_Last4** - Had weight loss of unknown cause in last 4 weeks
- **Had\_TIA** - Having TIA in last year
- **Has\_MI** - Having MI conditions

, in all previous waves before stroke occurred.

### Contingency table for each symptom

**Headache in last 4 months** 44 strokes out of a total of 327 strokes were found with headache in last 4 months, while 2138 no-strokes out of a total of 43542 no-strokes were found with headache in last 4 months.

Table 17: Contingency table of strokes with Headache in last 4 months

|                        | Stroke | No Stroke |
|------------------------|--------|-----------|
| <b>Symptom Present</b> | 44     | 2138      |
| <b>Symptom Absent</b>  | 283    | 41404     |

**Chest pain in last 4 months** 25 strokes out of a total of 327 strokes were found with chest pain in last 4 months, while 661 no-strokes out of a total of 43547 no-strokes were found with chest pain in last 4 months.

Table 19: Contingency table of strokes with Chest pain in last 4 months

|  | Stroke | No Stroke |
|--|--------|-----------|
|--|--------|-----------|

|                        | Stroke | No Stroke |
|------------------------|--------|-----------|
| <b>Symptom Present</b> | 25     | 661       |
| <b>Symptom Absent</b>  | 302    | 42886     |

**Exhaustion in last 4 months** 6 strokes out of a total of 327 strokes were found with exhaustion in last 4 months, while 126 no-strokes out of a total of 43561 no-strokes were found with exhaustion in last 4 months.

Table 21: Contingency table of strokes with Exhaustion in last 4 months

|                        | Stroke | No Stroke |
|------------------------|--------|-----------|
| <b>Symptom Present</b> | 6      | 126       |
| <b>Symptom Absent</b>  | 321    | 43435     |

**Unknown bleeding in last 4 months** 0 strokes out of a total of 327 strokes were found with unknown bleeding in last 4 months, while 8 no-strokes out of a total of 43561 no-strokes were found with unknown bleeding in last 4 months.

Table 23: Contingency table of strokes with Unknown bleeding in last 4 months

|                        | Stroke | No Stroke |
|------------------------|--------|-----------|
| <b>Symptom Present</b> | 0      | 8         |
| <b>Symptom Absent</b>  | 327    | 43553     |

**Unknown weight loss in last 4 months** 0 strokes out of a total of 327 strokes were found with unknown weight loss in last 4 months, while 4 no-strokes out of a total of 43561 no-strokes were found with unknown weight loss in last 4 months.

Table 25: Contingency table of strokes with Unknown weight loss in last 4 months

|                        | Stroke | No Stroke |
|------------------------|--------|-----------|
| <b>Symptom Present</b> | 0      | 4         |
| <b>Symptom Absent</b>  | 327    | 43557     |

**Had TIA** 23 strokes out of a total of 327 strokes were found with having TIA, while 0 no-strokes out of a total of 43561 no-strokes were found with having TIA.

Table 27: Contingency table of strokes with ever having TIA

|                        | Stroke | No Stroke |
|------------------------|--------|-----------|
| <b>Symptom Present</b> | 23     | 0         |
| <b>Symptom Absent</b>  | 304    | 43561     |

**Had MI** 24 strokes out of a total of 327 strokes were found with having MI, while 252 no-strokes out of a total of 43543 no-strokes were found with having MI.

Table 29: Contingency table of strokes with ever having MI

|  | Stroke | No Stroke |
|--|--------|-----------|
|--|--------|-----------|

|                        | Stroke | No Stroke |
|------------------------|--------|-----------|
| <b>Symptom Present</b> | 24     | 252       |
| <b>Symptom Absent</b>  | 303    | 43291     |

**Fisher's exact test for enrichment** Calculate odds ratio and  $p$ -value.

Table 31: Odds ratio and  $p$  values of stroke and no-stroke cases with each symptom

|                                      | Odds ratio | $p$                        | Stroke | Total at Stroke |
|--------------------------------------|------------|----------------------------|--------|-----------------|
| Had TIA                              | inf        | $5.237839 \times 10^{-50}$ | 23     | 327             |
| Has MI                               | 13.607104  | $1.501984 \times 10^{-18}$ | 24     | 327             |
| Exhaustion in last 4 months          | 6.443406   | $4.872607 \times 10^{-4}$  | 6      | 327             |
| Chest pain in last 4 months          | 5.370901   | $1.032002 \times 10^{-10}$ | 25     | 327             |
| Headache in last 4 months            | 3.010931   | $2.368314 \times 10^{-9}$  | 44     | 327             |
| Unknown bleed in last 4 months       | 0.000000   | $1.000000 \times 10^0$     | 0      | 327             |
| Unknown weight loss in last 4 months | 0.000000   | $1.000000 \times 10^0$     | 0      | 327             |

|                                      | No-Stroke | Total no-stroke |
|--------------------------------------|-----------|-----------------|
| Had TIA                              | 0         | 43561           |
| Has MI                               | 252       | 43543           |
| Exhaustion in last 4 months          | 126       | 43561           |
| Chest pain in last 4 months          | 661       | 43547           |
| Headache in last 4 months            | 2138      | 43542           |
| Unknown bleed in last 4 months       | 8         | 43561           |
| Unknown weight loss in last 4 months | 4         | 43561           |

**Kaplan-Meier analysis of survival probability** Define survive as being stroke-free. The Kaplan-Meier model fitted here estimates the probability of being free from stroke, if one of the symptoms mentioned earlier show up. The following tables show the possibility of surviving stroke-free after 22 years, in population with and without the symptoms.

Table 34: 22 year survival summary with Headache in last 4 months

| <b>Time</b> | Headache in last 4 months ( $n=1749$ ) | No headache in last 4 months ( $n=6075$ ) |
|-------------|----------------------------------------|-------------------------------------------|
| <b>0.0</b>  | 1.0000                                 | 1.0000                                    |
| <b>22.0</b> | 0.8962                                 | 0.9495                                    |

Table 36: 22 year survival summary with Chest pain in last 4 months

| <b>Time</b> | Chest pain in last 4 months ( $n=554$ ) | No chest pain in last 4 months ( $n=7270$ ) |
|-------------|-----------------------------------------|---------------------------------------------|
| <b>0.0</b>  | 1.0000                                  | 1.0000                                      |
| <b>22.0</b> | 0.8338                                  | 0.9419                                      |

Table 38: 22 year survival summary with Exhaustion in last 4 months

| <b>Time</b> | Exhaustion in last 4 months ( $n=130$ ) | No exhaustion in last 4 months ( $n=7694$ ) |
|-------------|-----------------------------------------|---------------------------------------------|
| <b>0.0</b>  | 1.0000                                  | 1.0000                                      |
| <b>22.0</b> | 0.8954                                  | 0.9353                                      |

Table 40: 22 year survival summary with Unknown bleeding in last 4 months

| <b>Time</b> | Unknown bleed in last 4 months ( $n=8$ ) | No unknown bleed in last 4 months ( $n=7816$ ) |
|-------------|------------------------------------------|------------------------------------------------|
| <b>0.0</b>  | 1.0000                                   | 1.0000                                         |
| <b>22.0</b> | 0.8571                                   | 0.9349                                         |

Table 42: 22 year survival summary with Unknown weight loss in last 4 months

| <b>Time</b> | Unknown weight loss ( $n=4$ ) | No unknown weight loss ( $n=7820$ ) |
|-------------|-------------------------------|-------------------------------------|
| <b>0.0</b>  | 1.0                           | 1.0                                 |

Table 44: 22 year survival summary with ever having TIA or MI in last 4 months

| <b>Time</b> | Had TIA ( $n=33$ ) | Never had TIA ( $n=7791$ ) | Had MI ( $n=228$ ) | Never had MI ( $n=7596$ ) |
|-------------|--------------------|----------------------------|--------------------|---------------------------|
| <b>0.0</b>  | 1.0                | 1.0000                     | 1.0000             | 1.0000                    |
| <b>22.0</b> | 0.0                | 0.9454                     | 0.6794             | 0.9426                    |

Stroke-free Survival Probability after occurrence of symptoms

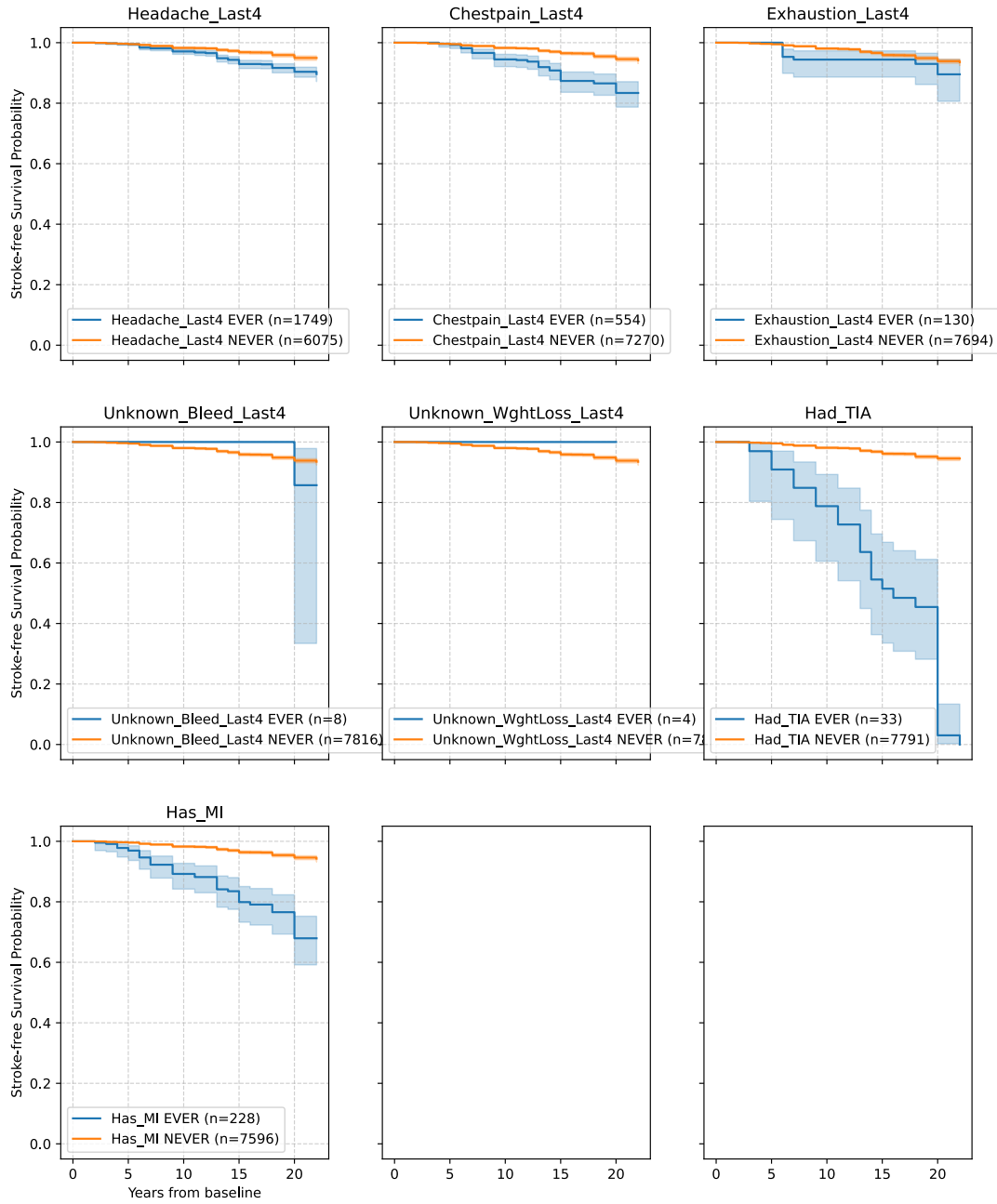


Figure 6: Stroke-free survival probability after occurrence of symptoms

From the results,

### Statistically significant symptoms associated with stroke

#### 1. Had TIA (odds $\gg 1$ , $p \ll 0.05$ )

Having TIA symptoms in 1-year window before survey is related very greatly with stroke risk in the subsequent wave. As shown by odds ratio being infinity, it means that all 21 individuals surveyed with TIA developed stroke in a subsequent wave. Statistically, this result is very significant, shown by the very low  $p$  value.

This agrees with existing literature about risk factors of stroke, where individuals having

TIA are extremely likely to develop stroke subsequently.

Survival probability decreases to 0% at 22 years. This is the highest rate of decrease of stroke-free population, meaning that ever having TIA means almost certainty of developing stroke.

2. **Having MI** (odds = 13.6071,  $p \ll 0.05$ )

Having MI before survey is related to more than 13 times higher stroke risk in the subsequent wave, and is statistically significant.

Survival function of it shows the survival probability decreased to 0.6794 at 22 years with exhaustion present, compared to 0.9426 with it absent. The difference is around 26%.

3. **Exhaustion** (odds = 6.4434,  $p < 0.05$ )

Exhaustion in 4-week window before survey is related to more than 6 times higher stroke risk in the subsequent wave, and is statistically significant.

Survival function of it shows the survival probability decreased to 0.8954 at 22 years with exhaustion present, compared to 0.9353 with it absent. The difference is around 4%.

However, exhaustion as a “symptom” is not a specific description. Also, only 6 individuals out of 304 with stroke experienced exhaustion in 4-month windows 1 wave before stroke. This result should be taken with a grain of salt.

4. **Chest pain** (odds = 5.3709,  $p \ll 0.05$ )

Chest pain in 4-week window before survey is related to more than 5 times higher stroke risk in the subsequent wave, and is statistically significant.

Its survival function shows the survival probability decreased to 0.8338 at 22 years with exhaustion present, compared to 0.9419 with it absent. The difference is around 10%, which is the greatest among all.

Clinically, chest pain may indicate cardiovascular issues. Some of which, like HT (under treatment), has been proven to be correlated with a higher stroke risk.

5. **Headache** (odds = 3.0109,  $p \ll 0.05$ )

Headache in 4-week window before survey is related to more than 3 times higher stroke risk in the subsequent wave, and is statistically significant.

Its survival function shows the survival probability decreased to 0.8692 at 22 years with exhaustion present, compared to 0.9299 with it absent. The difference is around 6%.

For 2 other symptoms, which are having bleeding of unknown cause and having weight loss of unknown cause, not enough data are available to support any claims to be made, as survival functions for cases where each symptom is present and absent are not complete.

Therefore, adding to the criteria for population with prioritized access to BP monitoring,

1. Population who have had a TIA or MI in recent years, and,
2. Population who shows symptoms of headache, chest pain, and exhaustion in recent years.

## 2.6 Evaluate criteria for monitoring scheme

In previous sections, the following criteria for prioritized access to BP monitoring schemes are found,

1. Population with higher DBP or SBP
2. Population with HT conditions or diabetes conditions (under treatment)
3. Population with higher BMI
4. Older population
5. Population who have had MI or TIA in recent years
6. Population who shows symptoms of headache, chest pain, and exhaustion in recent years.

To facilitate applying the criteria, a binary classification method based on determined cut-offs shall be used. Define positive as being susceptible to higher stroke risk, cut-offs for these criteria can be found to facilitate applying them.

This part finds optimal cut-offs for criteria, that satisfies the following,

- **Being easy to apply**



Have a clear cut-off with binary classification

- **Minimizes enforcement workload (WUD)**

The criteria should flag a manageable portion of the population as positive. This is judged from two factors, Work-up to detection (WUD) and population yield. Both should be kept at a manageable level.

WUD is defined as the number of people to flag and monitor, in order to catch one true positive case (having stroke). Mathematically, it is defined as

$$\text{WUD} = \frac{1}{\text{PPV}}$$

It can be seen that the smaller this value is, the smaller the population is needed to screen to detect true positive cases, and thus the smaller the workload to enforce such criteria.

- **Maximizes Positive Predictive Value (PPV)**

Positive Predictive Value (PPV) is defined as the probability that a person who tests positive (predicted higher stroke risk) is actually positive (develops stroke).

Mathematically, it is defined as,

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

TP = True positive, FP = false positive. In other words, it measures the rate of actual positives among all positive predictions.

- **Maximizes Sensitivity**

Higher sensitivity means a higher rate of predicting actual future strokes.

Mathematically, it is defined as,

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

TP = True positive, FN = false negative. In other words, it measures the rate of positives detected in all who later turned positive.

The following factors require finding optimal cut-offs,

1. Population with higher DBP or SBP (cut-off DBP and SBP measured in mmHg)
2. Population with higher BMI (cut-off BMI value)
3. Older population (cut-off age)
4. Population who have had a MI or TIA in recent years (cut-off time range to trace back)

**Note**

Since data about occurrence of symptoms like headache, chest pain and exhaustion are only recorded in 4-week windows before each survey, they may be inaccurate, as they may not accurately represent actual occurrence of such symptoms. Therefore, they are exempted from cutoff analysis.

**Strategy** This part will iterate through a range of cut-offs, and run a function to evaluate their performance. The following factors are considered,

- **Youden's index**, calculated by
  - **Sensitivity** - True positive rate
  - **Specificity** - True negative rate
- **Population yield** - Amount of population flagged positive.
- **WUD** - Work needed to detect a positive case.

Youden's index is defined as

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

**Helper functions for evaluating optimal cutoff value** This `evaluate_cutoff` function evaluates the effect on screening with the following factors

**Set ranges of test values** As age at baseline was filtered to be within 45-80 years of age, test values of age cutoff must be within this range as well. This ensures a meaningful comparison. The following graphs show the screening performances for a range of cutoffs for each variable.

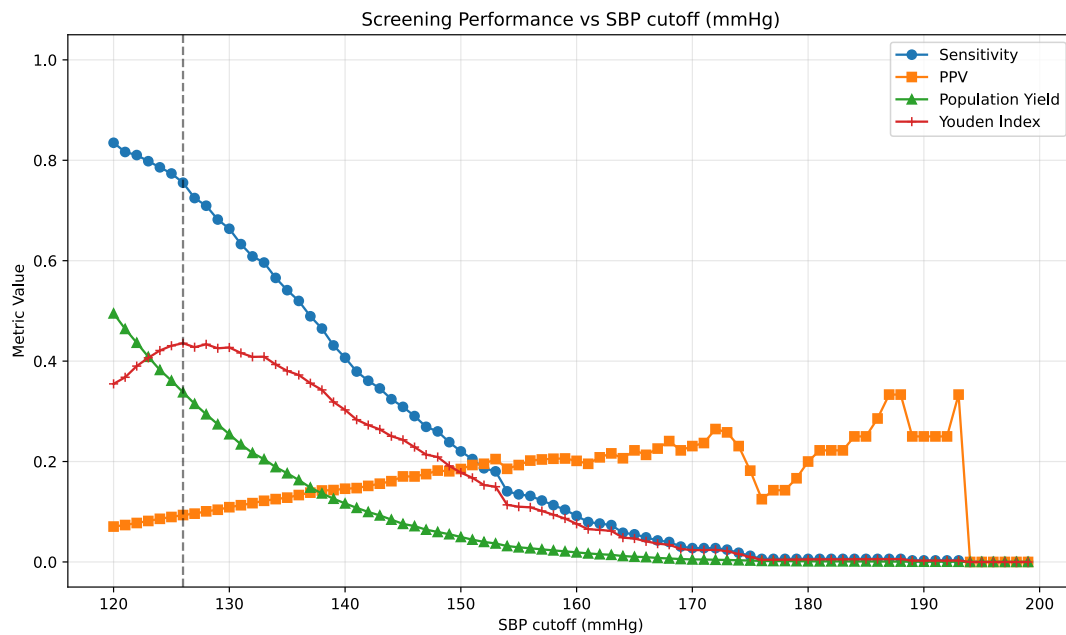


Figure 7: Screening performances for cutoffs of SBP

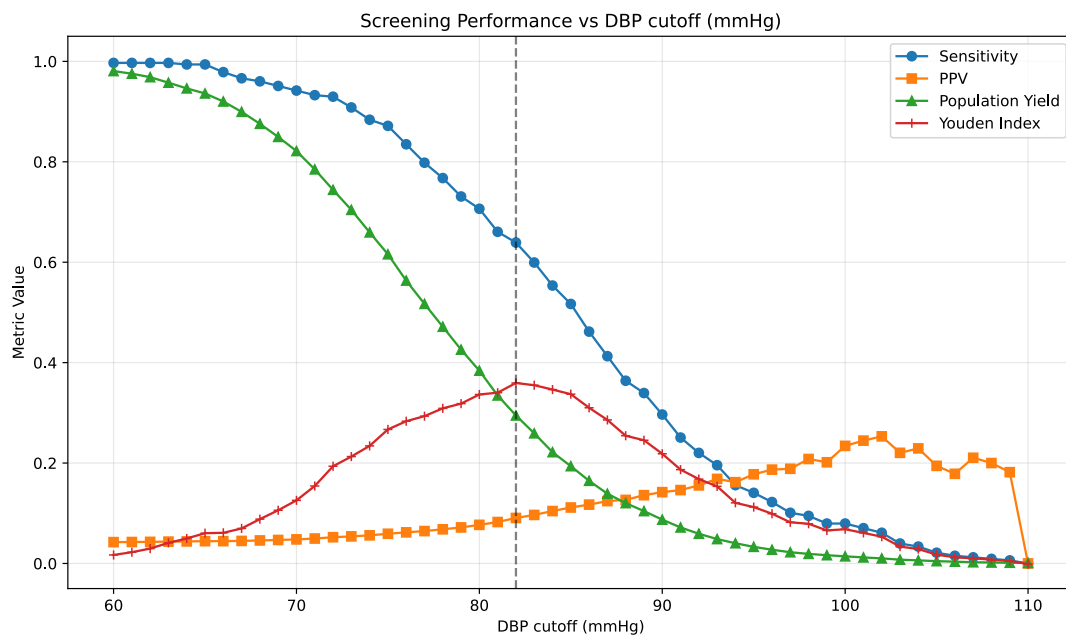


Figure 8: Screening performances for cutoffs of DBP

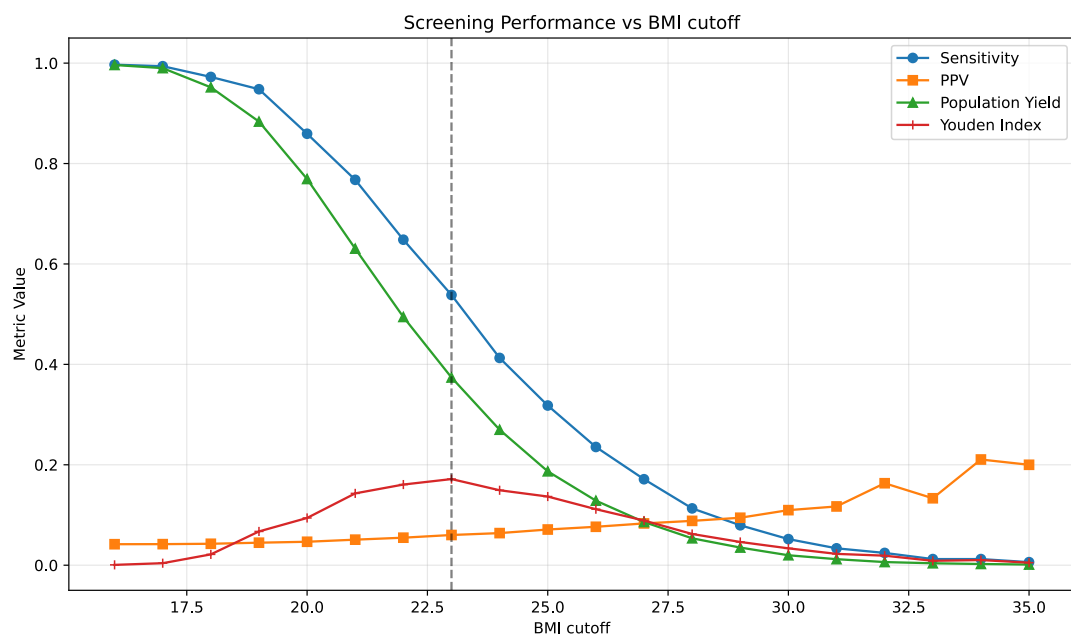


Figure 9: Screening performances for cutoffs of BMI

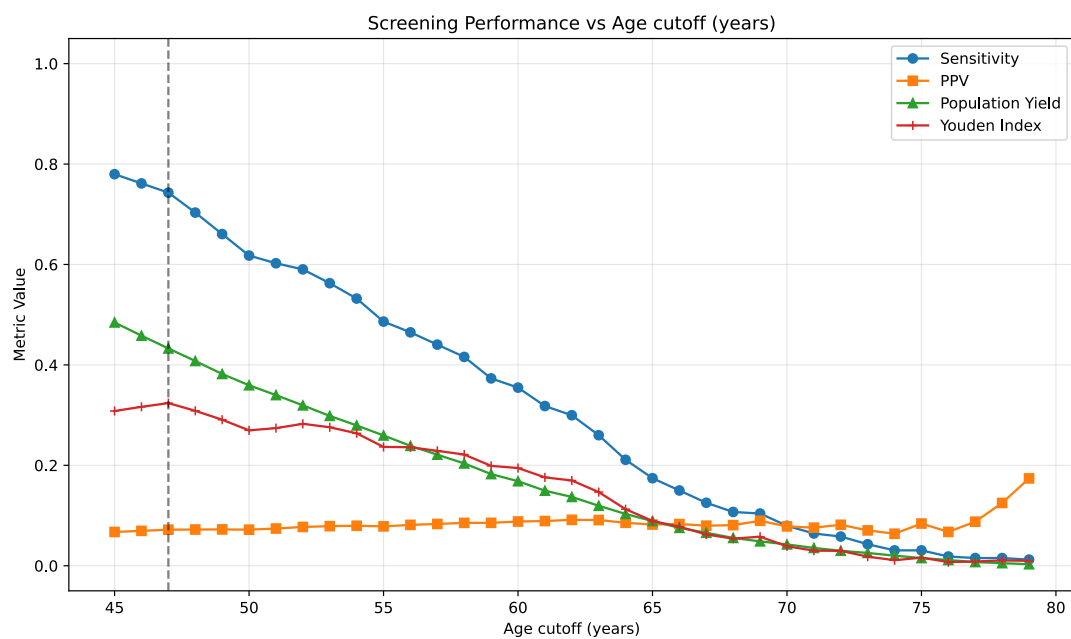


Figure 10: Screening performances for cutoffs of age

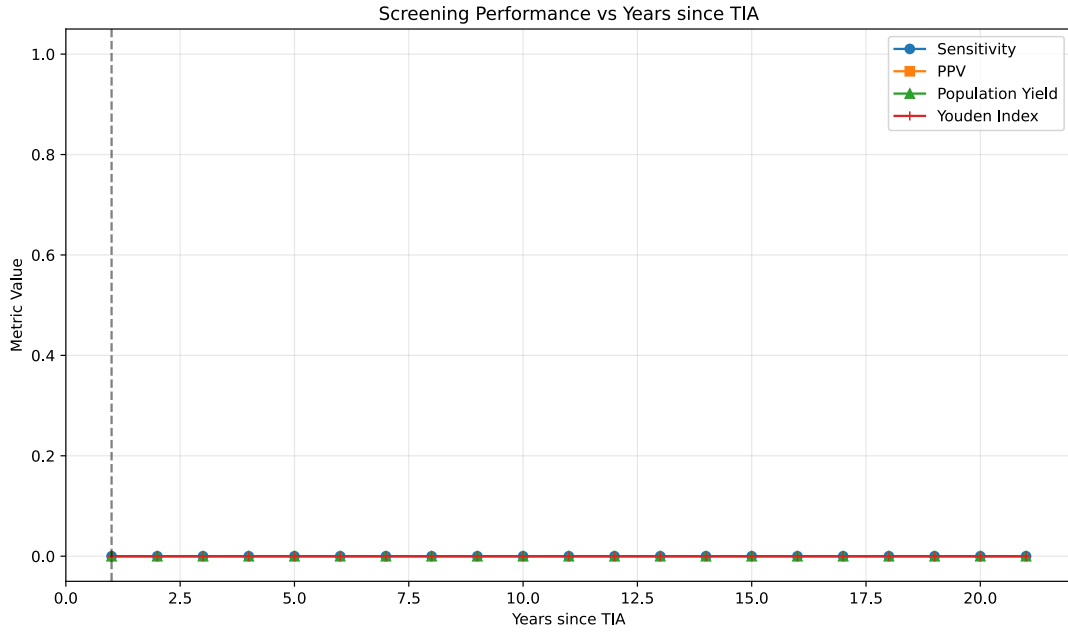


Figure 11: Screening performances for cutoffs of years since TIA

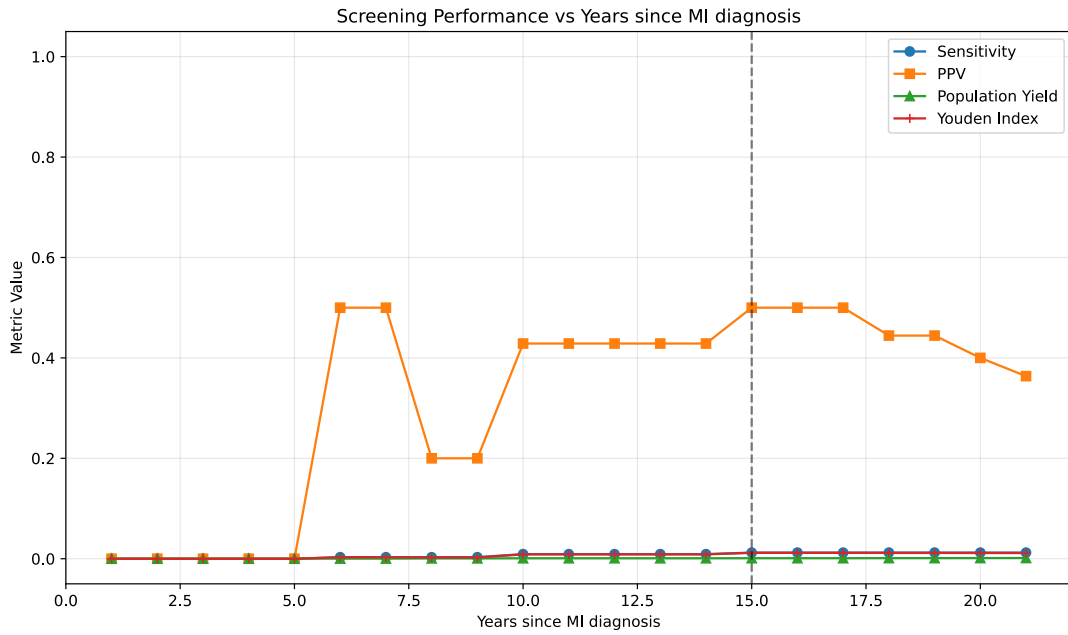


Figure 12: Screening performances for cutoffs of years since MI diagnosis

The following table summarizes optimal cutoffs found by maximizing Youden's index.

Table 46: Optimal cutoffs of each factor found by maximizing Youden's index

|             | SBP cutoff (mmHg) | DBP cutoff (mmHg) | BMI cutoff |
|-------------|-------------------|-------------------|------------|
| Cutoff      | 126.0000          | 82.0000           | 23.0000    |
| Sensitivity | 0.7554            | 0.6391            | 0.5382     |
| Specificity | 0.6803            | 0.7203            | 0.6333     |
| PPV         | 0.0934            | 0.0906            | 0.0602     |

|                  |         |         |         |
|------------------|---------|---------|---------|
| Population Yield | 0.3379  | 0.2947  | 0.3738  |
| Youden           | 0.4356  | 0.3594  | 0.1715  |
| WUD              | 10.7045 | 11.0335 | 16.6193 |

|                  | Age cutoff (years) | Years since TIA | Years since MI diagnosis |
|------------------|--------------------|-----------------|--------------------------|
| Cutoff           | 47.0000            | 1.0             | 15.0000                  |
| Sensitivity      | 0.7431             | 0.0             | 0.0122                   |
| Specificity      | 0.5808             | 1.0             | 0.9995                   |
| PPV              | 0.0718             | NaN             | 0.5000                   |
| Population Yield | 0.4328             | 0.0             | 0.0010                   |
| Youden           | 0.3239             | 0.0             | 0.0117                   |
| WUD              | 13.9342            | NaN             | 2.0000                   |

## Evaluation on cutoffs

**1. SBP and DBP** (SBP: WUD = 10.7045, Population yield = 0.3379, Sensitivity = 0.7554) (DBP: WUD = 11.0335, Population yield = 0.2947, Sensitivity = 0.6391)

By maximizing Youden's index, cutoff values are determined to be at higher than 82/130 mmHg. SBP and DBP at 126 mmHg and 82 mmHg respectively roughly match reference values of Stage 1 hypertension, which are 80/130 mmHg (American Heart Association, 2022). Around 10.7-11 people must be screened before one case of stroke can be predicted accurately, around 1/3 of the population would be flagged positive. Sensitivity is around 63-75%.

**2. BMI** (WUD = 16.6193, Population yield = 0.3738, Sensitivity = 0.5382)

By maximizing Youden's index, cutoff value is determined to be at higher than 23.0. Sensitivity is at around 53.82%. It flags around 37% of the population positive.

However, PPV does not increase much when cutoff values increase. The maximum is at around 0.2, decreasing WUD to 5 people. Nevertheless, BMI 23.0 is within normal range of BMI values, and sensitivity of this cutoff is only at 53%, such a cutoff value is not particularly significant in real life.

**3. Age** (WUD = 13.9342, Population yield = 0.4328, Sensitivity = 0.7431)

By maximizing Youden's index, cutoff value is determined to be at higher than 47 years at baseline. Sensitivity is at around 74%. Around 43% of the population is flagged positive. This matches roughly the "middle-aged" age group determined from existing research (Geifman et al., 2013), which is defined as 49-60 years of age.

## 4. Lookback Years since symptoms

**TIA** By maximizing Youden's index, cutoff lookback years is determined to be at 1 year. However, Youden's index is at 0.0, and PPV and WUD cannot be calculated normally. This shows that there has been errors in the computation, and this result should not be used. Even so, previous results show that occurrence of TIA has a very high correlation with stroke. Occurrence of TIA at any point should be treated with follow-up actions.

**MI** (WUD = 2.0000, Population yield = 0.0010, Sensitivity = 0.0122)

By maximizing Youden's index, cutoff lookback years is determined to be at 15 years. However, neither Youden's index nor PPV particularly increase as the cutoff value varies. This shows a

cutoff value matters little to monitoring schemes, as WUD therefore created varies little, and sensitivity does not significantly increase.

Therefore, this cutoff value is not particularly significant. Even so, previous results show that occurrence of MI has a very high correlation with stroke. Occurrence of MI at any point should be treated with follow-up actions.

**Conclusion / Criteria for prioritized BP monitoring scheme access** In conclusion, cutoffs for prioritized BP monitoring scheme access should be,

- **SBP/DBP** - 82/130 mmHg
- **Age** - Above 47 years
- **TIA and MI** - Diagnosed in the past

Other factors are - **HT** - Condition present, under treatment - **Diabetes** - Condition present, under treatment - **Symptoms** - Headache, chest pain, exhaustion present in the past

Population that match these criteria should get admitted to a BP monitoring scheme that continues to track occurrences / values of such factors, and new stroke risks are continuously evaluated over time. When necessary, intervention strategies against strokes shall be taken when stroke risk becomes high.

## 3 Proposed Health Policy

### 3.1 Policy Recommendation

Extend factors screened in CDCC and introduce a stroke risk screening program as an extension to the CDCC Pilot Scheme. In addition to existing screening for HT and diabetes in CDCC, monitoring for medical conditions such as TIA and MI and symptoms such as headache, chest pain and exhaustion can be added. Admit those who satisfy cutoff values found to a stroke monitoring program, where they receive proactive stroke prevention treatment alongside existing CDCC treatment for chronic diseases.

**Synthesis** As pointed out in literature review, stroke is related to a variety of factors, such as medical history of TIA, not just HT conditions (high BP). Analysis of data found also finds that treated HT still contributes to a higher stroke risk, and conditions like MI and symptoms such as headache, chest pain and exhaustion are related with stroke risk.

Therefore, only by using multifactorial monitoring can stroke risk be accurately identified and mediated in time, but current screening programs lacks screening for other factors found out to be related with stroke risk. If this were implemented, it would improve efficiency and effectiveness of CDCC in preventing strokes, supporting its mission of better health for people living with chronic diseases in Hong Kong.

**Implementation** The implementation of such scheme relies on an interconnected analytics platform, that stores screening data of participant such as BP measurements, medical history of conditions (TIA, MI) or symptoms of interest (headache, chest pain, exhaustion)

It can be better described using the Sense-Think-Act framework.

- **Sense** - Collect additional data alongside screening for HT and diabetes in CDCC
  - Medical history in past year (TIA and MI)
  - Symptoms in past year (headache, chest pain, exhaustion) and upload them to an interconnected analytics platform.

#### **Stakeholders involved**

- **CDCC participants** - Answer screening questions about medical history and symptoms, then receive measurements of BP and other checks in CDCC.

- **Family doctor assigned** - Follow screening procedures to measure BP and perform other checks in CDCC. Upload results to an analytics platform, update medical history on the platform as necessary.
- **Think** - Analyze results on analytics platform to determine stroke risk based on data on BP measurements, medical history, and symptoms.

**Stakeholders involved**

- **Platform managers** – Manage the analytics platform to ensure accurate detection of high-risk individuals.
- **Family doctor assigned** – Follow up with high-risk participants according to stroke management guidelines to plan follow-up with tailored preventive care against stroke.
- **Primary Healthcare Commission** - Establish and evaluate stroke management guidelines based on actual situations of Hong Kong population.
- **Act** - Initiate follow-up actions for those flagged high risk, and update stroke management guidelines as necessary

**Stakeholders involved**

- **Family doctor assigned** – Plan subsidized medical visits to follow-up with high-risk individuals, e.g., prescribe relevant medication.
- **CDCC participants** - Follow treatment plans and attend medical visits.
- **Platform managers** – update stroke management guidelines as necessary, e.g., adjust cut-offs for admission.

**Evaluation** To evaluate the implementation, short-term and long-term evaluations are needed. Short-term evaluation involve regular audits should be conducted to find ways to - Reduce false negatives and false positive rates - Increase detection rates of stroke - Increase cost-effectiveness by reducing WUD and population yield

Long-term evaluations should focus on reducing stroke occurrence rates and mortality within the screened population, compared to others.

Sustained improvements in above-mentioned outcomes show success of the policy.

### 3.2 Technology Description

An interconnected analytics platform with a mobile client should be integrated into the current CDCC scheme to better advanced risk screening and communication.

The platform stores screening data from participants, including BP readings, medical history of participants, including occurrences of TIA and MI, and symptoms such as headache, chest pain and exhaustion. When new data is uploaded, the platform automatically updates stroke risk scores of individual, and notifies family doctor assigned and participant about the result through automated notifications on the mobile client.

Both the family doctors assigned to participant through CDCC and the participant can view records and risk assessments through the mobile client. Through the mobile client, screening data and medical history can be updated by family doctors, while participants need to report recent symptoms frequently to ensure accuracy of stroke risk determined.

This interconnected platform enables timely interventions and keeps both patients and health-care providers actively informed and engaged.

**Value Proposition** This platform provides the following benefits,

1. **Allows early detection and intervention**

Since risk score is assessed automatically, family doctor assigned can formulate intervention strategies in time to prevent stroke if risk score is high. Delay is minimized as risk score is sent to the family doctor assigned instantly. This ensured early detection of stroke, so stroke risk of participant can be reduced directly.

## 2. **Fast and accurate calculation of stroke risk**

Since risk score is computed with automated algorithms, stroke risk is determined very quickly. It also eliminates delays associated with manual analysis and reduces the chances of human error, ensuring that both patients and doctors receive accurate risk assessments. Also, since the platform analyzes stroke risk from multiple factors, it can find high-risk individuals that may get missed by current CDCC screening with.

These make the platform more efficient in preventing stroke.

## 3. **Integrating seamlessly with CDCC**

The platform complements the scheme by incorporating new risk factors and automating the expanded screening process. To the participants, they now need to simply record their recent symptoms. To the family doctors, they now need to upload screening results and medical history to the platform. Both of which are small changes in procedures, which make the platform quick to adopt.

## 4. **Increase participation and awareness of participants**

The platform enables participants to report their recent symptoms directly on the mobile app, instead of during medical visits. This encourages them to become more aware of their health and risk factors of stroke. It also encourages self-monitoring of risk factors, increasing participation in stroke prevention. This means that participants would proactively cooperate with family doctors to prevent stroke.

**Feasibility** Integration of this platform may face the following challenges,

### 1. **Resistance to new technology from participants**

Since this approach needs participants to frequently record their recent symptoms to ensure accuracy of stroke risk determined, the aged population might find it hard to integrate this step into their routines, or even develop resistance to it.

**Solution** Offer in-person or online tutorials and support on using the platform mobile app. The app should also be designed with accessibility in mind. Periodically, user feedback should be evaluated to guide incremental improvements.

### 2. **Need training for family doctors**

Family doctors may be unfamiliar with the new analytics platform and hesitate to adapt routines to utilize it.

**Solution** Provide training sessions for family doctors in the CDCC with hands-on demonstration of using the platform. Distribute concise, written guides of using the platform for easy reference.

Training resources and guides should also be updated regularly to ensure that they are updated about new features in time.

### 3. **Concerns about data privacy**

Participants and staff may worry about unauthorized access or misuse of sensitive health data.

**Solution** Implement strong encryption and data security practices, and restrict data access to authorized users only. For example, limit screening data upload strictly to family doctor assigned to participant. No other parties may access the data. Also, clearly communicate security measures and privacy policies to all stakeholders.

In conclusion, the integration of the analytics platform with existing healthcare IT systems in CDCC can be done sustainably with clear education for all users (participants and their assigned family doctor), while continuously using user feedback to guide incremental improvements.



## AI usage note

Parts of this project was completed with the help of AI to,

- Refine language
- Suggest suitable data analysis methods
- Parse literature data found manually

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